

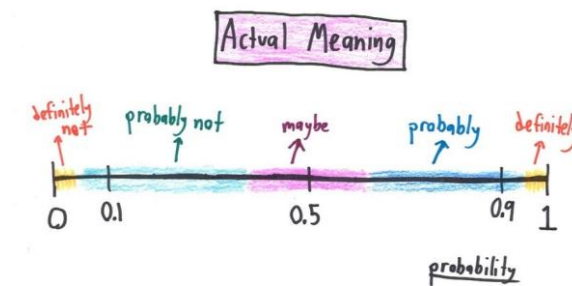


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 8 - Lecture No 16 – March 04, 2026



Announcements!

1. Office hours (Ramadan) – Tuesday & Thursday 11:30 to 12:30 PM
2. Midterm Examination?
3. Quiz No 05 – March 18, 2026
4. Assignment 2 – due March 8, 2026
5. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE – Spring 2026							
Course	Tutor	Email	Monday	Tuesday	Wednesday	Thursday	Friday
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	2:00-3:00		2:00-3:00		12:40-1:40
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	12:45-2:30		12:45-2:30		11:00-12:00
	Zain Naqi	zn09224@st.habib.edu.pk		9:00 - 10:30	11:45 - 12:45	9:00 - 10:30	

Agenda for today

1. Unit 4 – Discrete Random Variable

Functions of multiple random variables

$$Z = g(X, Y)$$

$$\rightarrow p_Z(z) = \mathbf{P}(Z = z) = \mathbf{P}(g(X, Y) = z) = \sum_{(x, y): g(x, y) = z} p_{X, Y}(x, y)$$

Expected value rule:

$$\rightarrow E[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X, Y}(x, y)$$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$\rightarrow E[X+Y]=E[g(x, y)]$$

$$\rightarrow g(x, y)=x+y$$

$$\rightarrow = \sum_x \sum_y (x+y) p_{X,Y}(x,y)$$

$$\rightarrow = \sum_x \sum_y (x) p_{X,Y}(x,y) + \sum_x \sum_y (y) p_{X,Y}(x,y)$$

$$\rightarrow = \sum_x x \sum_y p_{X,Y}(x,y) + \sum_y y \sum_x p_{X,Y}(x,y)$$

$$\rightarrow = \sum_x x p_X(x) + \sum_y y p_Y(y) = E[X] + E[Y]$$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$E[X_1 + \dots + X_n] = E[X_1] + \dots + E[X_n]$$

$$E[2X + 3Y - Z] = E[2X] + E[3Y] - E[Z] = 2E[X] + 3E[Y] - E[Z]$$

Conditional PMFs

$$A = \{ Y = y \}$$

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A)$$

$$p_{X|Y}(x \mid y) = \mathbf{P}(X = x \mid Y = y) = \frac{p(X = x, Y = y)}{p(Y = y)}$$

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \quad \text{defined for } y \text{ such that } p_Y(y) > 0$$

$$\sum_x p_{X|Y}(x|y) = 1$$

Y = Second roll

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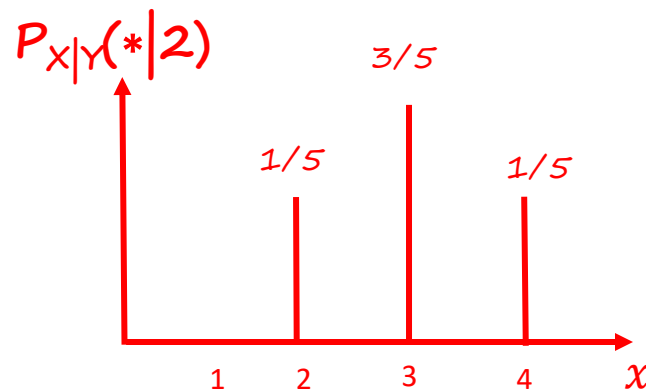
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll

$$\rightarrow P_Y(2) = 5/20$$

$$\rightarrow P_{X|Y}(1|2) = 0$$

$$\rightarrow P_{X|Y}(2|2) =$$



$$p_{X,Y}(x, y) = p_Y(y)p_{X|Y}(x \mid y)$$

$$p_{X,Y}(x, y) = p_X(x)p_{Y|X}(y \mid x)$$

Conditional PMFs involving more than two r.v.'s

- **Self-explanatory notation**

$$\begin{aligned}\rightarrow p_{X|Y,Z}(x|y,z) &= p(X=x \mid Y=y, Z=z) = \frac{p(X=x, Y=y, Z=z)}{p(Y=y, Z=z)} \\ &\rightarrow = \frac{p_{X,Y,Z}(x, y, z)}{p_{Y,Z}(y, z)}\end{aligned}$$

$$\rightarrow p_{X,Y|Z}(x, y|z) = p(X=x, Y=y \mid Z=z)$$

- **Multiplication rule**

$$P(A \cap B \cap C) = P(A) P(B \mid A) P(C \mid A \cap B)$$

$$A = \{X=x\} \quad B = \{Y=y\} \quad C = \{Z=z\}$$

$$p_{X,Y,Z}(x, y, z) = p_X(x) p_{Y|X}(y \mid x) p_{Z|X,Y}(z \mid x, y)$$

Conditional Expectation

$$A = \{ Y = y \}$$

$$\mathbf{E}[X] = \sum_x x p_X(x) \quad \mathbf{E}[X | A] = \sum_x x p_{X|A}(x) \quad \mathbf{E}[X | Y = y] = \sum_x x p_{X|Y}(x | y)$$

Expected Value Rule

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x) \quad \mathbf{E}[g(X) | A] = \sum_x g(x) p_{X|A}(x)$$

$$\mathbf{E}[g(X) | Y = y] = \sum_x g(x) p_{X|Y}(x | y)$$

Total probability and expectation theorems

$$Y = \{y_1, y_2, \dots, y_n\} \quad A_i = \{Y = y_i\}$$

- Partition of sample space in $A_1, A_2, \dots, A_n$
- $p_X(x) = P(A_1) p_{X|A_1}(x) + \dots + P(A_n) p_{X|A_n}(x)$

$$p_X(x) = \sum_y p_Y(y) p_{X|Y}(x|y)$$

- $\mathbf{E}[X] = P(A_1) \mathbf{E}[X | A_1] + \dots + P(A_n) \mathbf{E}[X | A_n]$

$$\mathbf{E}[X] = \sum_y p_Y(y) \mathbf{E}[X | Y = y]$$

- **Fine print:**

Also valid when Y is a discrete R.V. that ranges over an infinite set, as long as $\mathbf{E}[|X|] < \infty$

The mean of the binomial

- X : binomial with parameters n, p

❖ number of successes in n independent trials

$X_i = 1$ if i^{th} trial is a success; p
(indicator variable)
 $X_i = 0$ otherwise $1-p$

$$E[X] = np$$

$$X = X_1 + \dots + X_n$$

$$E[X] = \underbrace{E[X_1]}_p + \dots + \underbrace{E[X_n]}_p = np$$

$$E[X] = \sum_{k=0}^n k p_X(k) = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$